

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	1,590 bp	1,594 bp	1,587 bp
Protein Length	529 aa	431 aa	528 aa
Mutation Type	None	4-bp duplication / frameshift	3-bp deletion / in-frame deletion
Reading frame changed	No, Frame 1	Yes, frameshift	No, remains Frame 1
Premature stop codon appeared	No	Yes	No, expected
Amino acids affected	None	Begins at amino acid 427; downstream sequence is altered and protein is truncated	One amino acid deleted
Expected functional consequence	Normal HEXA function	Likely loss or major reduction of functional HEXA protein	Likely a small structural change; most of the protein sequence remains unchanged

This table shows the documented and artificial mutations had different effects on the HEXA protein. The documented mutation adds four nucleotides, causing a frameshift and premature stop codon, which makes the predicted protein much shorter. In contrast, the artificial mutation removes three nucleotides, so the reading frame stays the same and only one amino acid is expected to be removed. This shows that the number of nucleotides changed can affect how much a mutation changes the protein.